In ΔABC	In ΔDEF	In ΔGHI
$\sin\theta = \frac{3}{5}$	$\sin\theta = \frac{6}{10} = ?$	$\sin\theta = \frac{12}{20} = ?$
$\cos\theta = ?$	$\cos\theta = ?$	$\cos\theta = ?$
$\tan\theta = \frac{3}{4}$	$\tan\theta = ?$	$\tan\theta = ?$

Reciprocal ratios

We defined three basic trigonometric ratios namely, sine, cosine and tangent. The reciprocals of these ratios are also often useful during calculations. We define them as follows:

Basic Trigonometric Ratios	Its reciprocal	Short form
$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}}$	$\text{cosecant}\ \theta = \frac{\text{hypotenuse}}{\text{opposite side}}$	$\text{cosec}\ \theta = \frac{\text{hypotenuse}}{\text{opposite side}}$
$\cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$	$\text{secant}\ \theta = \frac{\text{hypotenuse}}{\text{adjacent side}}$	$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent side}}$
$\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}}$	$\text{cotangent}\ \theta = \frac{\text{adjacent side}}{\text{opposite side}}$	$\cot\theta = \frac{\text{adjacent side}}{\text{opposite side}}$

From the above ratios we can observe easily the following relations:

$$\text{cosec}\ \theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

$$\sin\theta = \frac{1}{\text{cosec}\,\theta} \qquad \cos\theta = \frac{1}{\sec\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

$(\sin\theta)\times(\text{cosec}\,\theta) = 1$. We usually write this as $\sin\theta\ \text{cosec}\,\theta = 1$.

$(\cos\theta)\times(\sec\theta) = 1$. We usually write this as $\cos\theta\ \sec\theta = 1$.

$(\tan\theta)\times(\cot\theta) = 1$. We usually write this as $\tan\theta\ \cot\theta = 1$.

Example 6.2 Find the six trigonometric ratios of the angle θ using the given diagram.

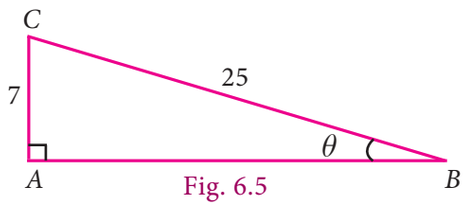


Fig. 6.5

Solution

By Pythagoras theorem,

$$AB = \sqrt{BC^2 - AC^2}$$
$$= \sqrt{(25)^2 - 7^2}$$
$$= \sqrt{625 - 49} = \sqrt{576} = 24$$

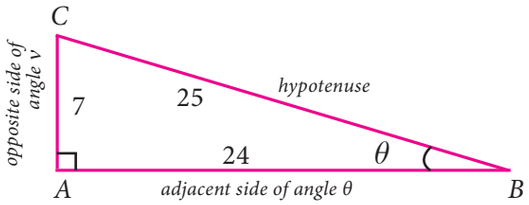


Fig. 6.6

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